

CBSE 2027 Chemistry — Predicted Paper (Set 2 of 10)

Based on the official CBSE Class 12 blueprint, historical PYQ trends, and the 2025-26 sample paper design [1-3], here is Set 2 of the highly probable predicted board paper. All questions are completely distinct from Set 1 and target high-frequency concepts [4-6].

SECTION A: Multiple Choice & Assertion-Reason Questions (1 Mark Each x 16 = 16 Marks)

Q1. The value of Henry's law constant (K_H) is: (a) greater for gases with higher solubility. (b) greater for gases with lower solubility. (c) constant for all gases at a given temperature. (d) independent of temperature. * **Chapter:** Solutions [7] | **Confidence:** Very High

Q2. How much charge, in terms of Faraday, is required for the reduction of 1 mole of MnO_4^- to Mn^{2+} ? (a) 1 F (b) 2 F (c) 3 F (d) 5 F * **Chapter:** Electrochemistry [8, 9] | **Confidence:** High

Q3. The unit of the rate constant (k) for a second-order reaction is: (a) s^{-1} (b) $mol \cdot L^{-1} \cdot s^{-1}$ (c) $L \cdot mol^{-1} \cdot s^{-1}$ (d) $L^2 \cdot mol^{-2} \cdot s^{-1}$ * **Chapter:** Chemical Kinetics [10, 11] | **Confidence:** Very High

Q4. Which of the following transition metals of the 3d series exhibits the maximum number of oxidation states? (a) Sc (b) Ti (c) Mn (d) Zn * **Chapter:** d- and f-Block Elements [12] | **Confidence:** Very High

Q5. What are the oxidation number and coordination number of Platinum in the complex $[Pt(en)_2Cl_2]^{2+}$? (a) +4 and 6 (b) +2 and 4 (c) +4 and 4 (d) +2 and 6 * **Chapter:** Coordination Compounds [13] | **Confidence:** High

Q6. Which of the following represents an allylic halide? (a) 1-Bromobutane (b) 1-Bromobut-2-ene (c) Bromobenzene (d) 1-Bromo-2-methylbenzene * **Chapter:** Haloalkanes and Haloarenes [14, 15] | **Confidence:** High

Q7. The major product formed when sodium ethoxide reacts with tert-butyl chloride is: (a) tert-butyl ethyl ether (b) 2-methylprop-1-ene (c) 1-butene (d) butane * **Chapter:** Alcohols, Phenols and Ethers [16, 17] | **Confidence:** Very High

Q8. Which of the following compounds will undergo a Cannizzaro reaction? (a) CH_3CHO (b) CH_3COCH_3 (c) C_6H_5CHO (d) CH_3CH_2CHO * **Chapter:** Aldehydes, Ketones and Carboxylic Acids [18, 19] | **Confidence:** Very High

Q9. Hinsberg's reagent, used to distinguish primary, secondary, and tertiary amines, is chemically: (a) Benzenesulphonyl chloride (b) Acetyl chloride (c) p-Toluenesulphonyl chloride (d) Chloroform and KOH * **Chapter:** Amines [20, 21] | **Confidence:** Very High

Q10. Which of the following nitrogenous bases is NOT present in DNA? (a) Adenine (b) Thymine (c) Cytosine (d) Uracil * **Chapter:** Biomolecules [22] | **Confidence:** High

Q11. The reaction of an alkyl chloride with NaI in dry acetone to form an alkyl iodide is known as the: (a) Swarts reaction (b) Wurtz reaction (c) Finkelstein reaction (d) Sandmeyer reaction * **Chapter:** Haloalkanes and Haloarenes [14, 23] | **Confidence:** Very High

Q12. Amino acids in a protein molecule are joined together by: (a) Glycosidic linkage (b) Peptide linkage (c) Phosphodiester linkage (d) Hydrogen bonds * **Chapter:** Biomolecules [22, 24] | **Confidence:** High

Directions for Q13 to Q16: Select the correct answer from the codes (a), (b), (c), and (d).

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A). (b) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (c) (A) is true, but (R) is false. (d) (A) is false, but (R) is true.

Q13. Assertion (A): Osmotic pressure is a colligative property. **Reason (R):** Osmotic pressure is directly proportional to the molarity of the solution at a given temperature. * **Chapter:** Solutions [25] | **Confidence:** High

Q14. Assertion (A): The molecularity of the elementary reaction $\text{H}_2 + \text{Br}_2 \rightarrow 2\text{HBr}$ appears to be 2. **Reason (R):** Two molecules of the reactants are involved in the given elementary reaction. * **Chapter:** Chemical Kinetics [26] | **Confidence:** Medium

Q15. Assertion (A): Ethanol has a higher boiling point than methoxymethane (dimethyl ether). **Reason (R):** Ethanol molecules are associated through intermolecular hydrogen bonding. * **Chapter:** Alcohols, Phenols and Ethers [27, 28] | **Confidence:** Very High

Q16. Assertion (A): Transition metals form alloys easily. **Reason (R):** Transition metals have similar atomic radii, allowing one metal atom to easily replace another in the crystal lattice. * **Chapter:** d- and f-Block Elements [29] | **Confidence:** High

SECTION B: Very Short Answer Questions (2 Marks Each x 5 = 10 Marks)

Q17. Define azeotropes. What type of azeotrope is formed by a mixture showing a strong negative deviation from Raoult's law? Give one example. * **Chapter:** Solutions [30, 31] | **Confidence:** Very High

Q18. Using Valence Bond Theory (VBT), explain the hybridization, geometry, and magnetic behaviour of the complex $[\text{Ni}(\text{CN})_4]^{2-}$. (Given atomic number of Ni = 28). * **Chapter:** Coordination Compounds [32] | **Confidence:** Very High

Q19. Give a simple chemical test to distinguish between the following pairs of compounds: (i) Methylamine and dimethylamine (ii) Aniline and N-methylaniline * **Chapter:** Amines [33, 34] | **Confidence:** High

Q20. Define a pseudo-first-order reaction. Give one suitable chemical example of such a reaction. * **Chapter:** Chemical Kinetics [35] | **Confidence:** High

Q21. Account for the following: (i) Copper(I) compounds are generally white/colourless whereas Copper(II) compounds are coloured. (ii) Ti^{4+} is diamagnetic while Ti^{3+} is paramagnetic in nature. * **Chapter:** d- and f-Block Elements [36, 37] | **Confidence:** Very High

SECTION C: Short Answer Questions (3 Marks Each x 7 = 21 Marks)

Q22. Calculate the osmotic pressure of a 5% (by mass) aqueous solution of urea (Molar mass = 60 g mol^{-1}) at 300 K. Assume the density of the solution is 1 g mL^{-1} . ($R = 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$). * **Chapter:** Solutions [38, 39] | **Confidence:** High

Q23. The conductivity of a 0.0024 M solution of acetic acid is $8.0 \times 10^{-5} \text{ S cm}^{-1}$. Calculate its molar conductivity (Λ_m) and degree of dissociation (α) if the limiting molar conductivity (Λ_m°) for acetic acid is $390.5 \text{ S cm}^2 \text{ mol}^{-1}$. * **Chapter:** Electrochemistry [40] | **Confidence:** Very High

Q24. A first-order reaction is 40% complete in 80 minutes. Calculate the value of the rate constant (k). In what time will the reaction be 90% completed? (Given: $\log 2 = 0.3010$, $\log 3 = 0.4771$, $\log 6 = 0.7782$). * **Chapter:** Chemical Kinetics [41, 42] | **Confidence:** Very High

Q25. How will you bring about the following organic conversions? (i) Ethanol to but-1-yne (ii) Chlorobenzene to p-nitrophenol (iii) 2-Bromopropane to 1-bromopropane * **Chapter:** Haloalkanes and Haloarenes [43, 44] | **Confidence:** High

Q26. Write the chemical equations involved in the following reactions: (i) Hydroboration-oxidation of propene. (ii) Reaction of phenol with concentrated HNO_3 in the presence of concentrated H_2SO_4 . (iii) Acid-catalyzed dehydration of ethanol at 413 K to form an ether. * **Chapter:** Alcohols, Phenols and Ethers [45, 46] | **Confidence:** Very High

Q27. Arrange the following sets of compounds in the increasing order of their indicated properties. Give a brief reason for each: (i) Acetaldehyde, Acetone, Methyl tert-butyl ketone (reactivity towards nucleophilic addition). (ii) Ethanol, Ethanoic acid, Benzoic acid (boiling point). (iii) 4-Nitrobenzoic acid, Benzoic acid, 4-Methoxybenzoic acid (acidic strength). * **Chapter:** Aldehydes, Ketones and Carboxylic Acids [17, 47] | **Confidence:** Very High

Q28. Answer the following: (i) What happens when D-glucose is treated with Bromine water? Write the equation. (ii) What structural change occurs when a native protein is subjected to a drastic change in pH or temperature? (iii) Name the specific linkage that connects monosaccharide units in starch. * **Chapter:** Biomolecules [22, 48] | **Confidence:** High

SECTION D: Case Study-Based Questions (4 Marks Each x 2 = 8 Marks)

Q29. Case Study 1: Batteries and Fuel Cells Galvanic cells that are designed to convert the energy of combustion of fuels like hydrogen, methane, or methanol directly into electrical energy are called fuel cells. One of the most successful fuel cells uses the reaction of hydrogen with oxygen to form water. The cell was used for providing electrical power in the Apollo space programme. The water vapors produced during the reaction were condensed and added to the drinking water supply for the astronauts. **(i)** Write the overall cell reaction that takes place in a Hydrogen-Oxygen fuel cell. (1 Mark) **(ii)** State one major advantage of

using a fuel cell over a conventional secondary battery. (1 Mark) **(iii)** If the standard Gibbs free energy of formation (ΔG°) of liquid water is $-237.1 \text{ kJ mol}^{-1}$, calculate the standard EMF (E°_{cell}) of the hydrogen-oxygen fuel cell. ($1F = 96500 \text{ C mol}^{-1}$). (2 Marks) * **Chapter:** Electrochemistry [49, 50] | **Confidence:** Very High

Q30. Case Study 2: Nucleophilic Substitution Mechanisms (S_N1 and S_N2) The reactivity of alkyl halides toward substitution depends heavily on the structure of the alkyl halide and the reaction conditions. The S_N2 mechanism involves the backside attack of a nucleophile, leading to the inversion of configuration. In contrast, the S_N1 mechanism proceeds via the formation of a planar carbocation intermediate, which is then attacked by the nucleophile from either side, often resulting in a racemic mixture if the starting material is chiral. **(i)** Out of 1-bromobutane and 2-bromobutane, which will react faster in an S_N2 reaction and why? (1 Mark) **(ii)** What is the expected stereochemical outcome (retention, inversion, or racemization) of an S_N1 reaction carried out on an optically active chiral alkyl halide? (1 Mark) **(iii)** A primary alkyl halide reacts with sodium alkoxide to form ethers via the Williamson synthesis, but when a tertiary alkyl halide is treated with a strong base like sodium methoxide, an elimination reaction predominates over substitution. Explain this observation. (2 Marks) * **Chapter:** Haloalkanes and Haloarenes [51, 52] | **Confidence:** High

SECTION E: Long Answer Questions (5 Marks Each x 3 = 15 Marks)

Q31. (A) d- and f-Block Elements (i) Describe the preparation of Potassium permanganate (KMnO_4) from pyrolusite ore (MnO_2). Write the balanced chemical equations for the two major steps involved. (3 Marks) (ii) Complete and balance the following ionic equation in an acidic medium: $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \rightarrow$ (2 Marks) * **OR (B)** Account for the following observations: (i) Actinoids show a much wider range of oxidation states compared to lanthanoids. (2 Marks) (ii) The lowest oxide of a transition metal is usually basic, whereas the highest oxide is often amphoteric or acidic. (2 Marks) (iii) E° value for the $\text{Mn}^{3+}/\text{Mn}^{2+}$ couple is much more positive than that for $\text{Cr}^{3+}/\text{Cr}^{2+}$. (1 Mark) * **Chapter:** d- and f-Block Elements [29, 53, 54] | **Confidence:** Very High

Q32. (A) Aldehydes, Ketones and Carboxylic Acids An organic compound (A) with molecular formula $\text{C}_9\text{H}_{10}\text{O}$ forms a yellow precipitate with 2,4-DNP reagent. It reduces Tollens' reagent and undergoes the Cannizzaro reaction. On vigorous oxidation with acidic KMnO_4 , it gives 1,2-benzenedicarboxylic acid (phthalic acid). Identify the compound (A) and write down all the chemical equations for the reactions involved (with 2,4-DNP, Tollens' reagent, Cannizzaro reaction, and oxidation). (5 Marks) * **OR (B)** Illustrate the following name reactions with one suitable chemical equation each: (i) Hell-Volhard-Zelinsky (HVZ) reaction (ii) Clemmensen reduction (iii) Etard reaction (iv) Decarboxylation of a carboxylic acid (v) Cross-aldol condensation * **Chapter:** Aldehydes, Ketones and Carboxylic Acids [55-57] | **Confidence:** Very High

Q33. (A) Chemical Kinetics (i) Derive the integrated rate equation for a zero-order reaction. Draw a graph of concentration of reactant $[\text{R}]$ versus time (t) for a zero-order reaction. (3

Marks) (ii) The rate of a reaction quadruples (becomes four times) when the temperature is raised from 293 K to 313 K. Calculate the energy of activation (E_a) of the reaction. ($R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$, $\log 4 = 0.6021$). (2 Marks) * **OR (B)** (i) Define the terms 'activation energy' and 'collision frequency' according to the collision theory of reaction rates. (2 Marks) (ii) A first-order reaction is 50% complete in 30 minutes at 300 K and 50% complete in 10 minutes at 320 K. Calculate the activation energy (E_a) for the reaction. ($R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$, $\log 3 = 0.4771$). (3 Marks) * **Chapter:** Chemical Kinetics [58, 59] | **Confidence:** Very High

CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.